Coding Set 3

1. Add Two Numbers Using Objects (Easy)

Create a class Number with one integer data member.

- Accept a value from the user.
- Create a non-member function add(Number, Number) that returns a Number object containing the sum.
- Display the result.

Concepts: Passing objects by value, returning objects.

---

2. Find Larger Object (Easy)

Create a class Student containing:

- Roll No
- Marks

Create a non-member function findTop(Student, Student) that returns the student having higher marks.

Concepts: Object comparison, returning objects.

---

3. Distance Addition (Easy-Medium)

Create a class Distance with:

- Feet
- Inches

Create a member function that accepts another Distance object as a parameter and returns the total distance after proper normalization.

Example:

5 ft 10 in

3 ft 8 in

----

9 ft 6 in

Concepts: Object as parameter to member function.

---

4. Bank Account Transfer (Medium)

Create a class BankAccount containing:

- Account Number
- Balance

Create a member function:

transfer(BankAccount &receiver, double amount)

which transfers money from one account to another.

Concepts: Passing objects by reference.

---

5. Complex Number Operations (Medium)

Create a class Complex.

Implement:

- Member function for addition.
- Non-member function for subtraction.
- Member function returning a new object for multiplication.

Concepts: Multiple styles of object passing.

---

6. Employee Salary Analysis (Medium)

Create a class Employee containing:

- Name
- Salary

Write:

- A non-member function that receives an array of Employee objects and returns the employee with highest salary.
- Another function that receives an Employee object and returns a revised Employee object with a 10% increment.

Concepts: Arrays of objects, returning objects.

---

7. Rectangle Comparison and Merge (Medium-Hard)

Create a class Rectangle with:

- Length
- Width

Write:

1. A member function that accepts another rectangle and checks whether both have equal area.

2. A non-member function that returns a new rectangle whose dimensions are the sums of corresponding dimensions.

Concepts: Object parameter passing, object return.

---

8. Library Book Exchange System (Hard)

Create a class Book with:

- Book ID
- Title
- Number of Copies

Create a member function:

exchange(Book &other)

that exchanges all information between two books.

Create another non-member function that receives two books and returns the one having more copies.

Concepts: Reference parameters, object manipulation.

---

9. Shopping Cart Billing System (Hard)

Create a class Product with:

- Product Name
- Price
- Quantity

Create:

- A non-member function that receives two Product objects and returns the product having higher total value (price × quantity).
- A member function that accepts another product and returns a new object representing their combined inventory.

Concepts: Returning objects, object composition.

---

10. University Result Processing System (Challenging)

Create a class Result containing:

- Roll Number
- Marks of 5 subjects

Write:

1. A member function that accepts another Result object and compares total marks.
2. A non-member function that receives three Result objects and returns the topper.
3. A function that receives a Result object and returns another Result object after applying grace marks (maximum 5 marks per subject, total grace not exceeding 20).

Concepts Covered:

- Passing objects to member functions
- Passing objects to non-member functions
- Returning objects
- Multiple object parameters
- Object comparison
- Real-world object manipulation